

ASSIGNMENT 4.1

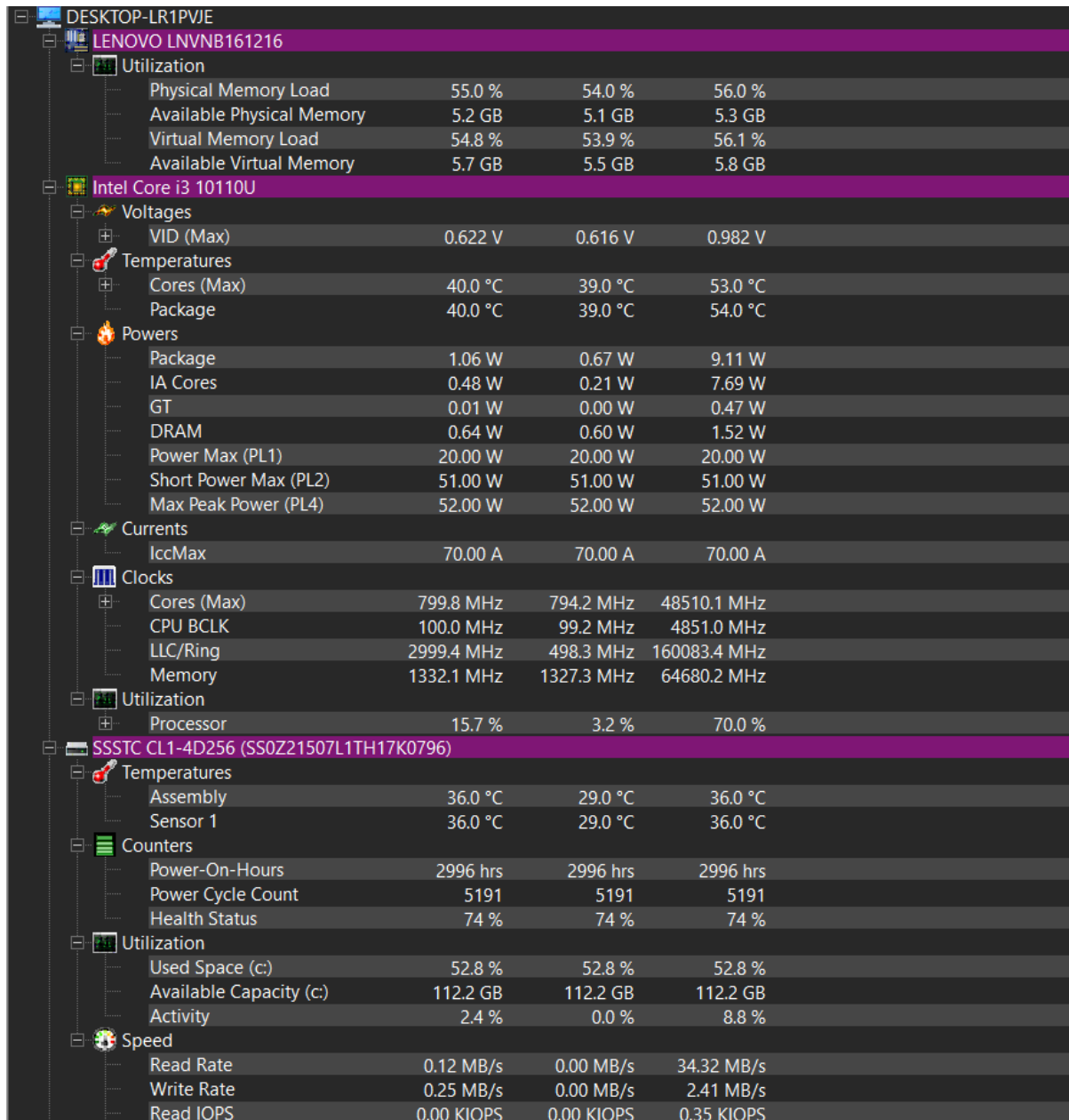
Q1. List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

Scenario	How the Multimeter Helps
1. Laptop not charging	It can measure the voltage from the laptop charger and charging circuit to check whether the correct voltage is reaching the laptop.
2. Desktop PC not turning on	It can check the power supply voltages to determine whether the PSU is providing proper power.
3. Smartphone not charging	It can be used to check voltage and continuity in the charging port or related circuit to identify a faulty connection or component.

Q2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video. Hint: Focus on the codes shown and how they relate to hardware issues

A POST card is used to diagnose hardware problems during the computer's startup process. It displays POST codes generated by the motherboard while checking hardware components. In the demonstration video, the POST card displayed a specific error code, which indicated a hardware-related problem. The code helped the technician identify the faulty component and troubleshoot the motherboard.

Q3. Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details. Constraint: Blur or crop out any personal information before submitting.



The screenshot displays the HWMonitor application interface. The left sidebar shows a tree view of system components: Utilization, Intel Core i3 10110U, and SSSTC CL1-4D256 (SS0Z21507L1TH17K0796). The main area shows detailed metrics for each component, organized into sub-sections like Voltages, Temperatures, Powers, Currents, Clocks, and Utilization. Each sub-section contains a table of values across three columns, likely representing different time intervals or sensors.

DESKTOP-LR1PVJE			
LENOVO LNVNB161216			
Utilization			
Physical Memory Load	55.0 %	54.0 %	56.0 %
Available Physical Memory	5.2 GB	5.1 GB	5.3 GB
Virtual Memory Load	54.8 %	53.9 %	56.1 %
Available Virtual Memory	5.7 GB	5.5 GB	5.8 GB
Intel Core i3 10110U			
Voltages			
VID (Max)	0.622 V	0.616 V	0.982 V
Temperatures			
Cores (Max)	40.0 °C	39.0 °C	53.0 °C
Package	40.0 °C	39.0 °C	54.0 °C
Powers			
Package	1.06 W	0.67 W	9.11 W
IA Cores	0.48 W	0.21 W	7.69 W
GT	0.01 W	0.00 W	0.47 W
DRAM	0.64 W	0.60 W	1.52 W
Power Max (PL1)	20.00 W	20.00 W	20.00 W
Short Power Max (PL2)	51.00 W	51.00 W	51.00 W
Max Peak Power (PL4)	52.00 W	52.00 W	52.00 W
Currents			
IccMax	70.00 A	70.00 A	70.00 A
Clocks			
Cores (Max)	799.8 MHz	794.2 MHz	48510.1 MHz
CPU BCLK	100.0 MHz	99.2 MHz	4851.0 MHz
LLC/Ring	2999.4 MHz	498.3 MHz	160083.4 MHz
Memory	1332.1 MHz	1327.3 MHz	64680.2 MHz
Utilization			
Processor	15.7 %	3.2 %	70.0 %
SSSTC CL1-4D256 (SS0Z21507L1TH17K0796)			
Temperatures			
Assembly	36.0 °C	29.0 °C	36.0 °C
Sensor 1	36.0 °C	29.0 °C	36.0 °C
Counters			
Power-On-Hours	2996 hrs	2996 hrs	2996 hrs
Power Cycle Count	5191	5191	5191
Health Status	74 %	74 %	74 %
Utilization			
Used Space (c:)	52.8 %	52.8 %	52.8 %
Available Capacity (c:)	112.2 GB	112.2 GB	112.2 GB
Activity	2.4 %	0.0 %	8.8 %
Speed			
Read Rate	0.12 MB/s	0.00 MB/s	34.32 MB/s
Write Rate	0.25 MB/s	0.00 MB/s	2.41 MB/s
Read IOPS	0.00 KIOPS	0.00 KIOPS	0.35 KIOPS

Q4. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	What it checks	Example Problem Detected
Multimeter	Checks voltage, resistance, and continuity in electrical circuits.	Faulty power supply or damaged charging circuit
POST Card	Checks the motherboard's startup process and displays POST error codes.	RAM, CPU, or motherboard hardware failure
Software Diagnostics	Checks hardware information, temperature, storage health, and system performance.	CPU overheating or failing HDD/SSD